

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 1

Student Name: T. Akshaya

Roll No.: A24126552122

Date:

1. TITLE

College Information Portal — HTML & HTML5 Fundamentals

2. AIM

To understand and implement the fundamental building blocks of HTML and HTML5 for creating structured web pages.

3. OBJECTIVE

- To create a valid HTML5 document using proper structure and semantic elements.
 - To apply text formatting, lists, tables, and images within a webpage.
 - To design a form using various input controls, including HTML5 input types.
 - To embed audio and video content in a webpage.
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4. THEORY

HTML5 provides semantic elements such as header, nav, section, article, and footer that describe the structure and meaning of a webpage, making it more organised and accessible. Text formatting tags, lists, tables, and images help present content clearly, while hyperlinks connect pages to other resources.

Forms allow users to enter and submit data using controls such as text boxes, radio buttons, checkboxes, dropdowns, and text areas. HTML5 also introduces new input types — such as date, number, color, and range — which provide built-in pickers and validation without extra scripting. Audio and video elements let media files be embedded directly into a webpage with native playback controls.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare `<!DOCTYPE html>` at the top to define it as an HTML5 document.
2. Add a `<header>` section containing the page heading and a `<nav>` with links to each section of the page.

3. Create an "About" <section> with an introductory paragraph, apply text formatting tags (bold, italic, underline, subscript, superscript), and add an ordered list, an unordered list, and hyperlinks.
 4. Create a "Students" <section> with a table displaying student details, followed by two images along with proper alt text.
 5. Create a "Register" <section> containing a form with text, email, password, radio button, checkbox, dropdown, textarea, and HTML5 input fields such as date, number, color, and range.
 6. Add Submit and Reset buttons at the end of the form.
 7. Create a "Media" <section> containing an <audio> element and a <video> element with playback controls.
 8. Add a <footer> section with copyright information.
 9. Save the HTML file and open it in a web browser to verify that the layout displays correctly.
 10. Test the navigation links, form fields, and media controls, and note down the results as test cases.
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6. TASK DESCRIPTION

A webpage titled "College Information Portal" was created, covering the following parts:

Part A: Basic HTML

- Created a valid HTML5 document.
- Displayed college information using headings and paragraphs.
- Applied text formatting (Bold, Italic, Underline, Superscript, Subscript).
- Created one ordered list (courses offered) and one unordered list (facilities available).
- Added hyperlinks to the college website and Google.

Part B: Tables and Images

- Created a student details table with Roll No, Name, Branch, and Email.
- Inserted the college logo and one relevant campus image with alternate text.

Part C: Forms

Designed a student registration form containing:

- Name, Email, Password
- Gender (Radio Buttons)
- Hobbies (Checkboxes)
- Branch (Dropdown)
- Comments (Textarea)
- Submit and Reset buttons

Part D: HTML5 Features

- Used semantic elements (header, nav, section, article, footer).
 - Embedded one audio and one video file.
 - Added HTML5 input types — Date, Number, Color, and Range.
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7. PROGRAM

[Task_1.html](#)

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>College Information Portal</title>
</head>
<body>

<!-- ===== PART D: Semantic Header & Nav ===== -
->
<header>
<h1>ANITS - College Information Portal</h1>
<nav>
  <a href="#about">About</a> |
  <a href="#students">Students</a> |
  <a href="#register">Register</a> |
  <a href="#media">Media</a>
</nav>
</header>

<!-- ===== PART A: Basic HTML ===== -->
<section id="about">
<h2>About the College</h2>
<article>
  <p>Anil Neerukonda Institute of Technology and Sciences (ANITS) is a
  premier engineering college offering courses in Computer Science,
  Electronics, Mechanical and more. It focuses on quality education,
  research and placements.</p>

  <h3>Text Formatting </h3>
  <p>
    <b>Bold Text</b> - Highlights important words.<br>
    <i>Italic Text</i> - Used for emphasis.<br>
    <u>Underlined Text</u> - Draws attention.<br>
    H<sub>2</sub>O (Subscript)<br>
    x<sup>2</sup> (Superscript)
  </p>

  <h3>Courses Offered (Ordered List)</h3>
  <ol>
    <li>CSE</li>
    <li>ECE</li>
    <li>CSM</li>
    <li>IT</li>
  </ol>

  <h3>Facilities Available (Unordered List)</h3>
  <ul>
    <li>Library</li>
    <li>Hostel</li>

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    <li>Sports Complex</li>
    <li>Wi-Fi Campus</li>
</ul>

<h3>Useful Links</h3>
<p>
    <a href="https://www.anits.edu.in" target="_blank">Visit College Website</a><br>
    <a href="https://www.google.com" target="_blank">Visit Google</a>
</p>
</article>
</section>

<!-- ===== PART B: Tables and Images ===== -->
<section id="students">
<h2>Student Details</h2>
<table border="1" cellpadding="8" cellspacing="0">
    <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
    </tr>
    <tr>
        <td>A24126552122</td>
        <td>Akshaya</td>
        <td>CSM</td>
        <td>teppala@anits.edu.in</td>
    </tr>
    <tr>
        <td>A241265520108</td>
        <td>Harshitha</td>
        <td>IT</td>
        <td>harshitha@anits.edu.in</td>
    </tr>
</table>

<h3>College Logo & Campus Image</h3>
 <br>

</section>

<!-- ===== PART C: Registration Form ===== -->
<section id="register">
<h2>Student Registration Form</h2>
<form action="#" method="post">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required><br><br>

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```
<label for="email">Email:</label>
<input type="email" id="email" name="email" required><br><br>

<label for="password">Password:</label>
<input type="password" id="password" name="password" required><br><br>

<label>Gender:</label>
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label><br><br>

<label>Hobbies:</label>
<input type="checkbox" id="reading" name="hobbies" value="reading">
<label for="reading">Reading</label>
<input type="checkbox" id="sports" name="hobbies" value="sports">
<label for="sports">Sports</label>
<input type="checkbox" id="coding" name="hobbies" value="coding">
<label for="coding">Coding</label><br><br>

<label for="branch">Branch:</label>
<select id="branch" name="branch">
  <option value="cse">CSE</option>
  <option value="ece">ECE</option>
  <option value="csm">CSM</option>
  <option value="it">IT</option>
</select><br><br>

<label for="comments">Comments:</label><br>
<textarea id="comments" name="comments" rows="4" cols="30"></textarea><br><br>

<!-- PART D: HTML5 Input Types -->
<label for="dob">Date of Birth:</label>
<input type="date" id="dob" name="dob"><br><br>

<label for="age">Age:</label>
<input type="number" id="age" name="age" min="17" max="30"><br><br>

<label for="favcolor">Favorite Color:</label>
<input type="color" id="favcolor" name="favcolor"><br><br>

<label for="rating">Satisfaction Level:</label>
<input type="range" id="rating" name="rating" min="0" max="10"><br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">
</form>
</section>

<!-- ===== PART D: Audio, Video, Iframe ===== -->
```

```

<section id="media">
  <h2>Media Section</h2>

  <h3>Audio</h3>
  <audio controls>
    <source src="college_anthem.mp3" type="audio/mpeg">
    Your browser does not support the audio tag.
  </audio>

  <h3>Video</h3>
  <video width="320" height="240" controls>
    <source src="campus_video.mp4" type="video/mp4">
    Your browser does not support the video tag.
  </video>

</section>

<!-- ===== Footer ===== -->
<footer>
  <p>&copy; 2026 ANITS College Information Portal. All Rights Reserved.</p>
</footer>

</body>
</html>

```

8. CODE EXPLANATION

Tag / Attribute	Purpose
<header>, <nav>, <section>, <footer>	HTML5 semantic tags used to structure the page into a header, navigation, content sections, and footer.
, <i>, <u>, <sub>, <sup>	Basic text formatting — bold, italic, underline, subscript, and superscript.
/	Ordered (numbered) list and unordered (bulleted) list.
	Creates a hyperlink that opens in a new tab.
<table>, <tr>, <th>, <td>	Creates the student details table with rows and columns.
	Displays an image, with alt text shown if the image fails to load.
<form>, <input>, <select>, <textarea>	Form and its input controls used to collect user data.
type="radio" / type="checkbox"	Radio allows one selection per group; checkbox allows multiple selections.
type="date" / "number" / "color" / "range"	HTML5 input types that give a date picker, numeric limit, colour picker, and slider.
<audio controls>, <video controls>	Embeds playable audio and video with built-in controls.

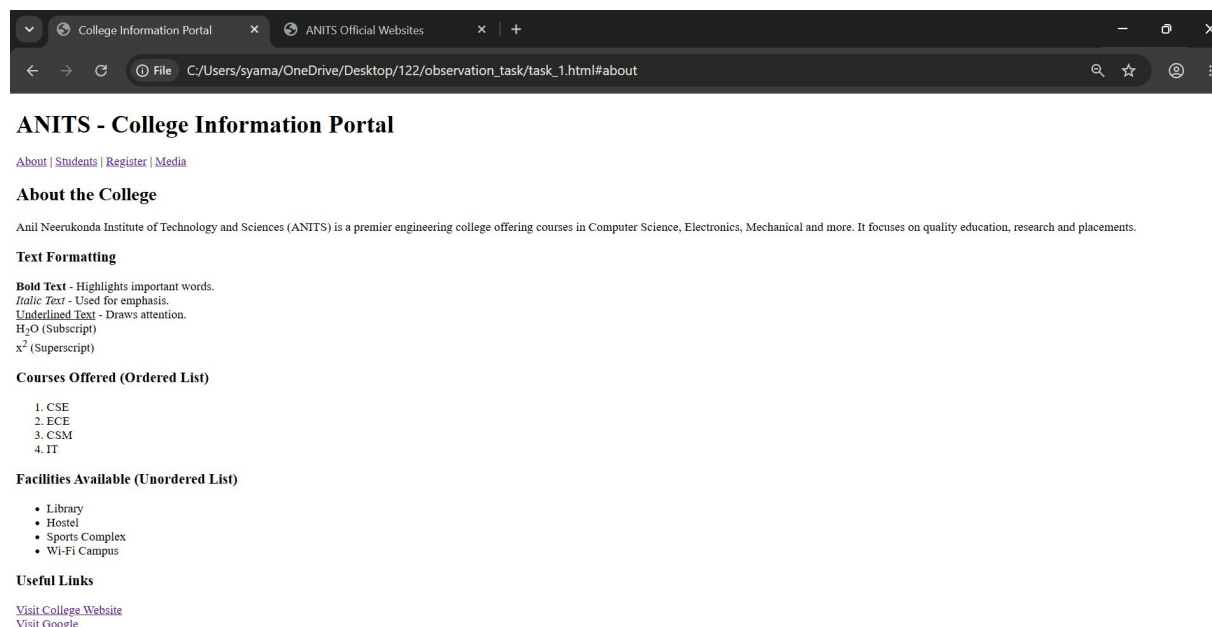
9. TEST CASES AND EXECUTION DATA

Test	Action	Expected Result	Actual Result	Status
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Case				
TC-01	Open the HTML file	All sections display correctly	Displayed correctly	Pass
TC-02	Click the website/Google links	Opens in a new tab	Opened in new tab	Pass
TC-03	Submit form with empty required fields	Shows validation error	Error shown, not submitted	Pass
TC-04	Select gender and hobbies	Radio allows one, checkbox allows many	Worked as expected	Pass
TC-05	Use date, number, color, range fields	Each shows its own control	All controls worked correctly	Pass

10. OUTPUT

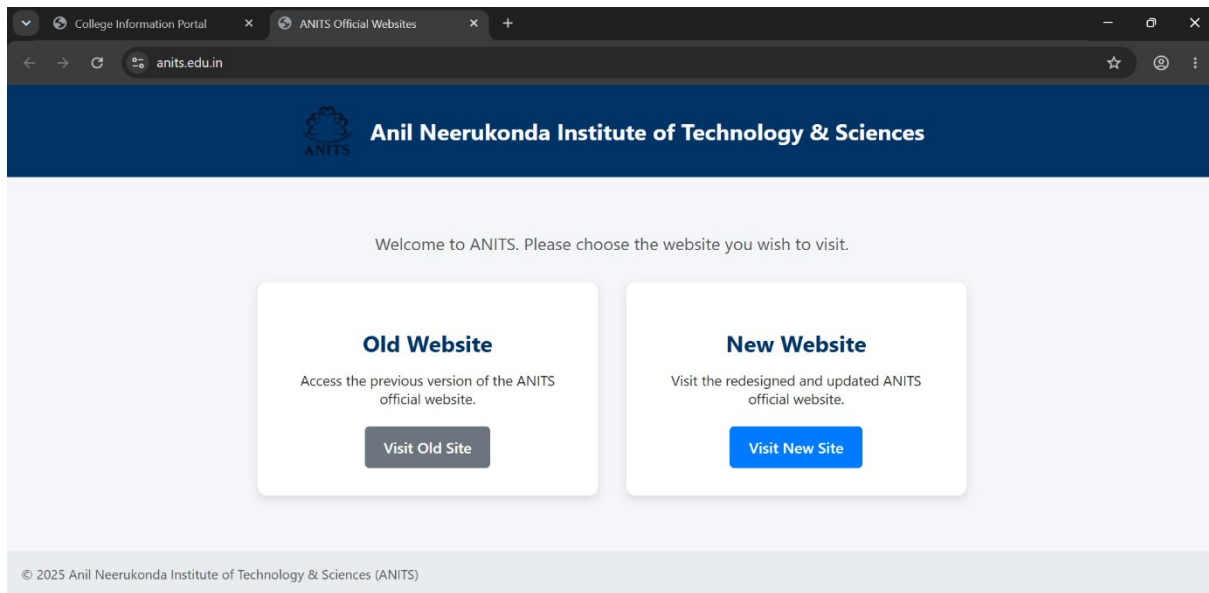
Output 1: About Section (Text Formatting & Lists)



Output 2: Click Website Links

Useful Links

[Visit College Website](#)
[Visit Google](#)



Output 3: Student Details Table & Images

Student Details

Roll No	Name	Branch	Email
A24126552122	Akshaya	CSM	teppala@anits.edu.in
A241265520108	Harshitha	IT	harshitha@anits.edu.in

College Logo & Campus Image



Output 4: Registration Form

Student Registration Form

Name:

Email:  Please fill out this field.

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Reading ☐ Sports ☐ Coding

Branch:

Comments:

Date of Birth: 

Age:

Favorite Color:

Satisfaction Level:

Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☒ Female

Hobbies: ☒ Reading ☐ Sports ☒ Coding

Branch:

Comments:

Date of Birth: 

Age:

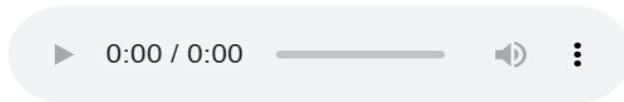
Favorite Color:

Satisfaction Level:

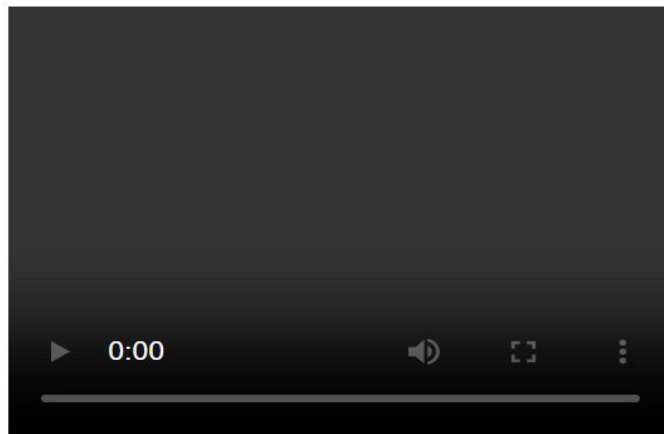
Output 5: Media Section (Audio & Video)

Media Section

Audio



Video



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11. RESULT

Thus, a webpage titled "College Information Portal" was successfully designed and developed using HTML and HTML5, covering basic tags, text formatting, lists, tables, images, forms, semantic elements, HTML5 input types, and audio/video embedding. The page was tested in a web browser and all features were verified to work correctly.